carbobydrates and lipids, they are not metabolised for providing energy. Normally, a well balanced diet will supply all the necessary vitamins in sufficient quantities.

Classification:

Vitamins are broadly classified into two groups based on their solubility.

i. Fat soluble vitamins

Vitamins A, D, E and K

ii. Water soluble vitamins

Vitamins B (B_1, B_2, B_3, B_6 & B_{12}), Biotin, Folic acid, Pantothenic acid and C

8.1 Fat soluble vitamins

The members of this group are vitamin A, vitamin D, vitamin E and vitamin K. All these vitamins are not soluble in water but are readily soluble in fat. These vitamins are present in the liver and fatty tissues. These vitamins can be stored in human body and hence getting too much of these vitamins can also cause problems.

8.1.1 Vitamin A (Retinol):

Vitamin A or Retinol is a poly isoprenoid compound containing a cyclohexenyl ring (Fig. 8.1) . It is found only in foods of animal origin. It is present in almost all species of fish, birds and mammals. The precursor of vitamin A, carotenoid is found in foods of plant origin. The body has the ability to convert carotenoid compound present in the diet into vitamin A. For example beta carotene found in vegetables such as carrot is converted into vitamin A by the symmetrical cleavege by the enzyme β-carotene -15,15'-dioxygenase.

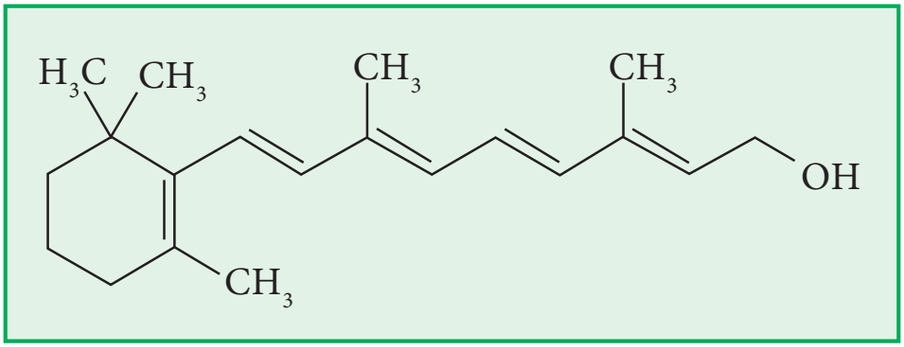

Figure 8.1 Structure of vitamin A

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